CHAPTER 3

CONTINUOUS PROBABILITY DISTRIBUTIONS

CHAPTER ESSENCE AT A GLANCE:
Master the Normal distribution, Standard Normal Variate (Z), bell-shaped curve properties, empirical area rules (68-95-99.7), and area probability calculations using Z tables.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Normal Distribution Definition:** Continuous distribution with parameters Mean (mu) and Variance (sigma^2); probability density function produces a symmetric, bell-shaped Gaussian curve asymptotic to the horizontal axis.

• **Standard Normal Variate (Z):** Standardizes any normal variable X via Z = (X - mu) divided by sigma; transformed distribution has Mean = 0, Variance = 1, and Standard Deviation = 1 [Z ~ N(0, 1)].

• **Geometric and Mathematical Properties:** Unimodal and perfectly symmetrical around mu; Mean = Median = Mode; points of inflection at mu - sigma and mu + sigma; Quartile Deviation approx 0.6745 sigma; Mean Deviation approx 0.8 sigma.

• **Empirical Area Rules:** Area under normal curve: mu +- 1 sigma encloses 68.27 percent of observations; mu +- 2 sigma encloses 95.44 percent; mu +- 3 sigma encloses 99.73 percent (virtually all data).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad [-\infty < x < \infty]$$

Normal Probability Density Function Formulation

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1) \rightarrow \text{Standard Normal Transformation}$$

Zero-Mean Unit-Variance Standardization Variable

$$P(\mu \pm \sigma) = 0.6827, \quad P(\mu \pm 2\sigma) = 0.9544, \quad P(\mu \pm 3\sigma) = 0.9973$$

Empirical Gaussian Probability Bounds

$$\text{Quartile and Mean Deviation} \Longleftrightarrow \text{QD} \approx \tfrac{2}{3}\sigma \quad \text{and} \quad \text{MD} \approx \tfrac{4}{5}\sigma$$

Approximation Ratios Linking Dispersion to Standard Deviation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The normal curve approaches the horizontal axis asymptotically from -infinity to +infinity and NEVER touches or intersects the axis.

• **Alert:** Total area under the normal curve is strictly equal to 1; due to symmetry, the area to the left of Z = 0 is 0.5 and to the right of Z = 0 is 0.5.

• **Alert:** In formula card 3, remember that 95 percent of data lies within mu +- 1.96 sigma, while 99 percent lies within mu +- 2.58 sigma.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State any four important properties of the Normal Distribution curve. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Bell-Shaped and Symmetrical:** The curve is perfectly symmetrical about the vertical line X = mu, where Mean = Median = Mode. **[1 Mark]**

• **2. Asymptotic to X-Axis:** The two tails extend indefinitely to -infinity and +infinity without ever touching the horizontal axis. **[1 Mark]**

• **3. Points of Inflection:** The curve changes its curvature from concave to convex at distances X = mu - sigma and X = mu + sigma. **[1 Mark]**

• **4. Total Area and Kurtosis:** Total area under the curve is 1 (50 percent on either side of mean); it is mesokurtic with Beta2 = 3 and Skewness = 0. **[1 Mark]**

2. The monthly income of a group of workers follows normal distribution with Mean = Rs 10,000 and Standard Deviation = Rs 2,000. Find the percentage of workers whose income is: (a) Greater than Rs 12,000, (b) Less than Rs 8,000 (given area between Z=0 and Z=1 is 0.3413). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Parameters:** mu = 10,000, sigma = 2,000. Transformation Z = (X - mu) divided by sigma. **[1 Mark]**

• **(a) Greater than 12,000 P(X > 12,000):** Z = (12,000 - 10,000) divided by 2,000 = +1.0. Area = 0.5 - Area(0 to 1) = 0.5 - 0.3413 = 0.1587 (15.87 percent). **[1.5 Marks]**

• **(b) Less than 8,000 P(X < 8,000):** Z = (8,000 - 10,000) divided by 2,000 = -1.0. By symmetry, Area = 0.5 - Area(0 to 1) = 0.5 - 0.3413 = 0.1587 (15.87 percent). **[1.5 Marks]**

3. Define Standard Normal Distribution. What are its Mean, Standard Deviation, and Variance? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** A special case of the normal distribution where the random variable is transformed into a standard score Z = (X - mu) divided by sigma. **[1.5 Marks]**

• **Parameters:** Mean = 0, Standard Deviation = 1, and Variance = 1. Notation: Z ~ N(0, 1). **[1.5 Marks]**

4. For a normal distribution, if the Standard Deviation is 15, calculate: (a) Quartile Deviation, (b) Mean Deviation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Quartile Deviation:** Formula is QD = (2 divided by 3) * sigma = (2 divided by 3) * 15 = 10 (or 0.6745 * 15 = 10.12). **[1.5 Marks]**

• **(b) Mean Deviation:** Formula is MD = (4 divided by 5) * sigma = (4 divided by 5) * 15 = 12 (or 0.7979 * 15 = 11.97). **[1.5 Marks]**

5. What percentage of total area under a normal curve lies between: (a) mu - 2 sigma and mu + 2 sigma, (b) mu - 3 sigma and mu + 3 sigma? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) mu +- 2 sigma:** Exactly 95.44 percent of total area (approximately 95.4 percent). **[1 Mark]**

• **(b) mu +- 3 sigma:** Exactly 99.73 percent of total area (virtually all observations). **[1 Mark]**

6. What are the points of inflection of a normal distribution with Mean = 50 and Standard Deviation = 8? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Formula:** Points of inflection are at X = mu - sigma and X = mu + sigma. **[1 Mark]**

• **Calculation:** X1 = 50 - 8 = 42 and X2 = 50 + 8 = 58. **[1 Mark]**